

SALEH DARZI

Applied and Post-Quantum Cryptography Researcher

 salehdarzi@usf.edu

 8135779609

 Tampa, Florida, USA

 www.salehdarzi.com

 scholar.google.com/salehdarzi

 saleh-darzi-4212b591

 TheSalehDarzi

 0000-0002-3465-1173

 Visa Status: F-1, I-485 Pending

EXPERTISE & RESEARCH INTERESTS:

**Applied Cryptography Researcher**

**Post-Quantum Cryptography Specialist**

**Network Security Engineer**

**Privacy-Preserving Machine Learning**

**Designing Security Frameworks**
for Wireless Networks, Artificial Intelligence (AI), Internet-of-Things (IoT), and Blockchain

EDUCATION

Ph.D. in Computer Science and Engineering

University of South Florida, Tampa, Florida

 Dec. 2021 – May 2027 (expected)

Thesis title: Post-Quantum Secure Authentication and Privacy-Preserving Frameworks for Next-Generation Wireless Networks
Applied Cryptography Research Laboratory (ACRL), Bellini College of Artificial Intelligence, Cybersecurity, and Computing. Advisor: Prof. Attila A. Yavuz.

M.Sc. Electrical Engineering (Communication-System)

K. N. Toosi University of Technology, Tehran, Iran

 Sept. 2017 – Dec. 2020

Thesis title: Improving Privacy-Preserving Techniques for Smart Grid using Lattice-based Cryptography
Developed **LPM2DA** (lattice-based privacy-preserving data aggregation) and **SPDBlock** (blockchain-based secure and privacy-preserving smart-grid framework).

B.Sc. Electrical Engineering (Electronic)

Islamic Azad University of Central Tehran Branch, Tehran, Iran

 Sept. 2013 – Jan. 2017

Thesis title: Cognitive Radio: Analysis of Spectrum Sensing in Cognitive Radio Networks

SKILLS

Languages & DevOps: C C++ Python Java Scripting Bash Git MATLAB x86 Assembly Ansible Rust Go Latex HTML MySQL Docker Prometheus

Libraries & Tools: OpenSSL Lib-OQS HELib NTL (Number Theory Library) GMP LibTomCrypt Libsodium (NaCl) Crypto++ libpqcrypto SageMath Microsoft SEAL PALISADE OpenFHE ZK-SNARK PBC WolfSSL MIRACL ecc-cpp Libschnorr pqm4

EXPERIENCE

Graduate Research Associate

University of South Florida

 May 2024 – Ongoing

- Funded By:** NSF–SNSF, Cisco Systems
- Serving as lead researcher, author, and security framework designer on NSF-funded projects, including the international SATUQ collaboration with ZHAW and the 5G authentication project with Purdue University, UT Dallas, and Muma College of Business. Designed and evaluated **post-quantum secure, scalable, and resilient cybersecurity frameworks** for hybrid satellite–terrestrial and next-generation wireless networks.
- Conducted extensive research and development across a broad spectrum of **cryptographic primitives**, including **post-quantum digital signatures, identity-based and group signatures, ring signatures, anonymous credentials, private information retrieval, post-quantum Tor, proof-of-work and time-lock puzzles, key encapsulation mechanisms, and distance-bounding protocols**, utilizing multiple cryptographic libraries for secure design, implementation, and benchmarking.

System Administrator and Technical Support

📅 April 2024 – Jan 2025

USF Bellini College of Artificial Intelligence, Cybersecurity and Computing

- Implemented enterprise security and compliance measures by deploying **Rapid7 agents** across multi-platform systems (Windows, Red Hat, Ubuntu, macOS) and automating **CIS Level 1 & 2 validation** through **Ansible playbooks**.
- Administered the **GAIVI computing cluster**, resolving capacity and node failures, developing diagnostic automation scripts, and performing on-site data-center maintenance and server deployments.
- Created a **centralized audit and configuration repository** to log, document, and remotely monitor all departmental servers and workstations.
- Designed and maintained a **Grafana-based monitoring dashboard** for real-time visualization of network and system performance across the department.
- Automated routine IT operations by developing **Ansible scripts** for remote configuration, updates, and digital signage control, reducing manual workload and downtime.
- Installed and maintained hardware components (**GPUs, memory, printers**) across labs and data centers, ensuring optimal performance and reliability.
- Configured **network protocols, user environments, and software setups** for faculty and research staff, ensuring secure, stable, and consistent access across heterogeneous systems.

Graduate Research Assistant

📅 December 2021 – May 2024

University of South Florida

- Lead researcher and security framework designer on multiple NSF- and industry-funded projects, including the **SaTC** initiative on secure distributed computing for 5G/6G networks, the **CAREER** project on post-quantum lightweight authentication for IoT, and the **Cisco Research Award** project on privacy-preserving and resilient cybersecurity frameworks for next-generation networks.
- Conducted research, design, and implementation of diverse **advanced cryptographic constructions**, including **post-quantum digital signatures, fully homomorphic encryption, secure multi-party computation, zero-knowledge proofs, anonymous credentials, private information retrieval, Counter-DoS mechanisms, privacy-preserving machine learning, and key encapsulation schemes**, employing multiple cryptographic libraries for development and performance evaluation.

Graduate Research Assistant

📅 Jan 2019 – Dec 2021

K. N. Toosi University of Technology

- Served as a **research investigator** in the **National Smart Metering Program (Iran)**, focusing on privacy-preserving techniques for smart grids using lattice-based cryptography.
- Authored 2 papers and led the design and implementation of **fully homomorphic encryption (FHE)** and **post-quantum digital signature schemes**, focusing on security, privacy, and performance optimization of smart grid networks.

Instructor & Graduate Teaching Assistant

📅 January 2019 – Present

University of South Florida & K. N. Toosi University of Technology

- **Courses:** CIS 4930/6930 Cryptography: Theory and Practice CIS4200: Penetration Testing for IT
CIS 4212/6214: Privacy-Preserving and Trustworthy Cyber-Infrastructures Computer Logic Design Lab
Lattice-Based Cryptography Information Theory Network Security
- Served as **instructor and lecturer** for advanced cryptography and cybersecurity courses, covering **PQC, lattice-based cryptography, and ECC**.
- Designed and delivered **hands-on cryptographic exercises, coding projects, and practical labs** in C/C++ and Python using libraries such as **OpenSSL, LibTomCrypt, LibOQS, NTL, GMP**, and others.

Internship

📅 June 2017 – December 2017

Noor Tab Tavan Technical and Engineering Company (Iran)

- Served as an **Electronic Engineer and Project Supervisor** for an **electrical grid modernization project**, overseeing system design, implementation, and performance testing.

SELECTED PUBLICATIONS

Patents

- Attila A. Yavuz, **Saleh Darzi**, "System and Method for Secure Location-Proof and Anonymous Privacy-Preserving Spectrum Access" (**SLAP**), U.S. Patent Application US 2025/0392478 A1; USF TTO ref. 24T238PR-CS, Provisional Filed: 06/16/2025.
- Attila A. Yavuz, **Saleh Darzi**, "Resilient Authentication for Next-Generation Wireless Networks with Lawful Interception and Quantum-Safe Forgery Detection," USF TTO ref. 25T059PR-CS, Provisional Filed: 11/22/2024.
- Attila A. Yavuz, **Saleh Darzi**, "System and Method for Privacy-Preserving and Post-Quantum Secure Counter Denial of Service for Spectrum Management in Next-Generation Wireless Networks" (**PACDoSQ**), U.S. Patent Application US 2025/0392468 A1; USF TTO ref. 24T238PR-CS, Provisional Filed: 06/21/2024.

Conferences

- **Saleh Darzi**, Gökcan Cantali, Dina Abdelhadi, Attila A. Yavuz, Gürkan Gür, "**QLAP**: A Quantum-Resilient Security and Privacy Framework for Dynamic Spectrum Access," in **MobiArch 2026: The 21st ACM Workshop on Mobility in the Evolving Internet Architecture**, in conjunction with ACM MobiCom 2026, Austin, TX, USA. (*Accepted; camera-ready submitted.*)
- **Saleh Darzi**, Attila A. Yavuz, "Privacy-Preserving and Post-Quantum Counter Denial of Service Framework for Wireless Networks" (**PACDoSQ**), in MILCOM 2024 – 2024 IEEE Military Communications Conference (MILCOM), pp. 1076–1081, 2024.
- **Saleh Darzi**, Attila A. Yavuz, "**SLAP**: Secure Location-Proof and Anonymous Privacy-Preserving Spectrum Access," in IEEE SVCC 2025: Silicon Valley Cybersecurity Conference, 2025.
- **Saleh Darzi**, Kiarash Sedghighadikolaie, Saif E. Nouma, Attila A. Yavuz, "Trustworthy AI Systems Through Lenses of Post-Quantum Security and Privacy-Enhancing Techniques," in CyberBay 2025: Tampa Bay Center of Excellence for Cybersecurity & AI, 2025.
- **Saleh Darzi**, Attila A. Yavuz, "Counter Denial of Service for Next-Generation Networks within the Artificial Intelligence and Post-Quantum Era," in the 2024 IEEE 6th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA), pp. 138–147, 2024.
- Attila A. Yavuz, Kiarash Sedghighadikolaie, **Saleh Darzi**, Saif E. Nouma, "Beyond Basic Trust: Envisioning the Future of NextGen Networked Systems and Digital Signatures," in the 2023 5th IEEE International Conference on Trust, Privacy and Security in Intelligent Systems and Applications (TPS-ISA), pp. 267–276, 2023.

Under Review:

- **Saleh Darzi**, Mirza Masfiquir Rahman, Imtiaz Karim, Rouzbeh Behnia, Attila A. Yavuz, Elisa Bertino, "A Lightweight Post-Quantum Authentication Framework for 5G Base Station Bootstrapping," under review at **NDSS Symposium 2027**, Seoul, Republic of Korea.
- Dina Abdelhadi, Gökcan Cantali, **Saleh Darzi**, Attila A. Yavuz, Gürkan Gür, "Public Key Infrastructure for UAVs over 5G NR Sidelink in the Post-Quantum Era," under review at the 17th IFIP Wireless and Mobile Networking Conference (WMNC), 2026.

Journals

- **Saleh Darzi**, Gökcan Cantali, Attila A. Yavuz, Gürkan Gür, "Privacy-Preserving and Secure Spectrum Sharing for Database-Driven Cognitive Radio Networks" (**SLAPX**), in Elsevier Computer Networks (COMNET) Journal, vol. 286, pp. 112420, issn. 1389-1286, 2026.
- **Saleh Darzi**, Kiarash Sedghighadikolaie, Saif E. Nouma, Attila A. Yavuz, "**QPADL**: Post-Quantum Private Spectrum Access with Verified Location and DoS Resilience," in Elsevier Computer Networks (COMNET) Journal, 2026. (*Accepted; to appear.*) Preprint: arXiv:2510.03631.
- Attila A. Yavuz, **Saleh Darzi**, Saif E. Nouma, "**LiteQSign**: Lightweight and Quantum-Safe Signatures for Heterogeneous IoT Applications," in IEEE Access, 2025.
- **Saleh Darzi**, Bahareh Akhbari, Hassan Khodaiemehr, "**LPM2DA**: A Lattice-Based Privacy-Preserving Multi-Functional and Multi-Dimensional Data Aggregation Scheme for Smart Grid," in Springer Cluster Computing Journal, vol. 25, no. 1, pp. 263–278, 2022.

Under Review:

- **Saleh Darzi**, Mirza Masfiquir Rahman, Imtiaz Karim, Rouzbeh Behnia, Attila A. Yavuz, Elisa Bertino, "Future-Proofing Authentication Against Insecure Bootstrapping for 5G Networks: Feasibility, Resiliency, and Accountability" (**BORG**), Elsevier Computer Networks (COMNET) Journal. *Second-round revision*. Preprint: arXiv:2510.23457.
- Gökcan Cantali, Dina Abdelhadi, Gürkan Gür, **Saleh Darzi**, Attila A. Yavuz, "Quantum-Safe Orbital Data Centers and AI: Use Cases, Challenges, and Research Directions," under review at the Special Issue on Orbital Edge: Data Centers in Space, 2026.

E-Prints

- **Saleh Darzi**, Attila A. Yavuz, Rouzbeh Behnia, "Post-Quantum Security for Trustworthy Artificial Intelligence: An Emerging Frontier," in Authorea Preprints, 2024.
- **Saleh Darzi**, Kasra Ahmadi, Saeed Aghapour, Attila A. Yavuz, Mehran Mozaffari Kermani, "Envisioning the Future of Cyber Security in the Post-Quantum Era: A Survey on PQ Standardization, Applications, Challenges and Opportunities," in arXiv preprint arXiv:2310.12037, 2023.

SERVICES & ENGAGEMENT

Program Committee (PC) & Chair:

- Publication Chair of the ACM CCS Workshop on Quantum-Resistant Cryptography and Security (QRSEC 2025), part of the ACM CCS 2025 Organizing Committee, Taipei, Taiwan.
- Program Committee of the IEEE Silicon Valley Cybersecurity Conference (SVCC 2025, SVCC 2026)

Reviewer:

- Reviewer of ACM Conference on Computer and Communications Security (ACM CCS-2025); Reviewer of USENIX Security Symposium (2025, 2026); Reviewer of IEEE Transactions on Dependable and Secure Computing Journal (TDSC); Reviewer of IEEE Transactions on Information Forensics and Security Journal (TIFS); Reviewer of ACM Computing Surveys Journal; Reviewer of IEEE Transactions on Smart Grid Journal; Reviewer of Annual Computer Security Applications Conference (ACSAC); Reviewer of IEEE IoT Journal; Reviewer of Journal of Information Security and Applications; Reviewer of Cluster Computing Journal; Reviewer of Concurrency and Computation: Practice and Experience

Invited Talks & Conference Presentations:

- Presented research papers at major venues, including "IEEE MILCOM 2024", "Silicon Valley Cybersecurity Conference (SVCC 2025)", "IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (IEEE TPS-ISA 2024)", and other international cryptography and network security conferences.
- Delivered the *Major Area Presentation*, "Post-Quantum Secure Authentication and Privacy-Preserving Frameworks for Next-Generation Wireless Networks," University of South Florida, November 10, 2025.
- Delivered a public research talk on my patents concerning post-quantum security for spectrum access systems at the *Innovation Match: Become the Founder of a USF Spinout*, organized by the University of South Florida Technology Transfer Office.
- Presentation of "Trustworthy AI Systems Through Lenses of Post-Quantum Security and Privacy-Enhancing Techniques" at CyberBay 2025, Tampa, Florida.

Awards & Grants:

- Recipient of multiple travel grants and research support awards for international conferences and NSF-funded projects, including NSF travel grants for MILCOM and IEEE TPS-ISA, and USF Research and Travel Support Grants.

Open-Source Contributions:

- Developer and maintainer of several **open-source cryptography projects** hosted on GitHub, including post-quantum authentication for 5G and IoT (**BORG-Scheme**) and GPU-accelerated private information retrieval and proof-of-work generation for post-quantum spectrum access (**QPADL**), along with other related repositories.

Professional Memberships:

- Member, **Institute of Electrical and Electronics Engineers (IEEE)**, **IEEE Communications Society**, **IEEE Young Professionals**, **Association for Computing Machinery (ACM)**, **Iranian Society of Cryptology**.

REFERENCES

Prof. Attila A. Yavuz

University of South Florida, Tampa, FL, 33620

@ attilaayavuz@usf.edu

☎ +1 (813) 974 0419

Prof. Mehran Mozaffari-Kermani

University of South Florida, Tampa, FL, 33620

@ mehran2@usf.edu

☎ +1 (813) 974 2671

Prof. Reza Azarderakhsh

Florida Atlantic University, Boca Raton, FL, 33431

@ razarderakhsh@fau.edu

☎ +1 (561) 297-4980

Prof. Rouzbeh Behnia

University of South Florida, Tampa, FL, 33620

@ behnia@usf.edu

☎ +1 (941) 359-4605